

Day – 2 Introduction to dynamic routing theory

Task - 1 Difference between static and dynamic routing.

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Routing	Static Routing	Dynamic Routing
Route Configuration	Manually configured by a network administrator.	Discovered and calculated automatically using routing protocols (e.g., OSPF, BGP, RIP).
Adaptability & Failover	Cannot adapt to network changes; broken links require manual updates.	Automatically reroutes traffic around link or node failures in real time.
Resource Usage	Zero CPU/RAM usage for route updates; no control traffic bandwidth overhead.	Consumes CPU, memory, and link bandwidth to calculate paths and exchange routing tables.
Network Scalability	Poor scalability; management complexity increases exponentially with network size.	High scalability; easily handles large, expanding networks with minimal extra configuration.
Security	Higher baseline security since paths are fixed and immune to malicious route injections.	Lower baseline security; vulnerable to rogue route advertisements if not properly authenticated.
Route Algorithm	None (hardcoded static paths).	Uses complex algorithms (e.g., Dijkstra's shortest path, Distance Vector) to find best paths.

Task – 2 Make pre – config network with three routers.

- Displays active routing protocols, update timers, hop count limits, and connected routing neighbors. Displays the active routing table, showing route sources, administrative distances, metrics (hop count), and next-hop IP addresses.

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- 1.Open Packet Tracer and Access CLI:Router 1 Setup.
- Open Cisco Packet Tracer and open your pre-configured 3-router file.
- Click on Router 1 (e.g., R1) and navigate to the CLI tab.
- Press Enter to access the command prompt.

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- 2.Verify RIP Status on Router 1:Command Execution.
- Run the following command to check RIP settings and updates:

```
R1> enable  
R1# show ip protocols
```

- Key outputs to observe and note down:
- Routing Protocol: Confirms Routing Protocol is "rip".
- Sending/Receiving Updates: Verifies update timers (Default: 30 seconds).
- Routing Information Sources: Lists neighboring router IPs providing updates.

- 3.Inspect RIP Routes on Router 1:Command Execution.
- Run the following command to inspect the routing table:

```
R1# show ip route
```

- Key outputs to observe and note down:
- Locate routes marked with the code R (which stands for RIP).
 - o 120 = Default Administrative Distance for RIP.
 - o X = Metric (Hop Count to the target subnet).
- Record the target network, the next-hop IP address (via x.x.x.x), and the outgoing interface.

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- 4.Repeat Verification on Routers 2 and 3:Routers 2 & 3 Setup.
- Repeat the exact same sequence on Router 2 and Router 3:

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```
- R2# show ip protocols
R2# show ip route

R3# show ip protocols
R3# show ip route
```

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Router Name	Route Code (R)	Target Network / Subnet Mask	Admin Distance	Next-Hop IP (via ...)	Outgoing Interface
Router 1	R	<i>e.g., 192.168.3.0/24</i>	[120/2]	10.0.0.2	Serial0/1/0
Router 2	R	<i>e.g., 192.168.1.0/24</i>	[120/1]	10.0.0.1	Serial0/1/0
Router 3	R	<i>e.g., 192.168.1.0/24</i>	[120/2]	10.0.0.5	Serial0/1/1

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Task – 3 Network change by shutting down the system.

- Procedure to Simulate Network Failure & Measure Convergence Time
- To observe convergence in RIP, you can perform a link failure test by shutting down an interface.
- Step 1: Shut Down an Interface on Router 1
- Open the CLI of Router 1 and administratively disable the interface connected to the LAN network (or the serial link to Router 2):

```
Router1> enable
Router1# configure terminal
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# shutdown
Router1(config-if)# exit
```

- Step 2: Monitor Routing Tables on Remaining Routers
- Immediately switch to the CLI of Router 2 (or Router 3) and run show ip route repeatedly every few seconds:
- Router2# show ip route
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- Scenario A: Triggered Updates Active (Immediate Convergence ~0–30 Seconds)
- Immediate Reaction: When Router 1 detects local link failure, it immediately sends a Poisoned Route update with a metric of 16 (Unreachable) to its neighbors. Running show ip route on Router 2 shows the route marked as possibly down or deleted within a few seconds.
- Scenario B: Timer-Based Convergence (~240 Seconds / 4 Minutes)
- If updates do not reach neighboring routers immediately, RIP relies on its default internal timers:

Elapsed Time	RIP Timer State	Observed Status in show ip route on Router 2 / 3
0 – 30s	Update Timer	Route remains active; metric looks normal ([120/1]).
180s (3 Min)	Invalid Timer Expires	The route is flagged as invalid and marked as possibly down in the routing table.
180s – 240s	Holddown Timer	The router suppresses higher-metric route advertisements for this subnet.
240s (4 Min)	Flush Timer Expires	The unreachable route is completely removed/flushed from the routing table.

Task – 4 How dynamic routing.

- Dynamic routing in delivery apps like Uber or Zomato mirrors RIP in computer networks by using continuous, automated updates to recalculate the optimal path from origin to destination without human intervention. Just as RIP routers periodically broadcast local traffic metrics (hop counts) to adjacent neighbors, delivery platforms collect live GPS and traffic updates to re-evaluate road network

conditions. When an interface fails or a road is blocked, both systems automatically discard the stale path and propagate a newly calculated route across the system. This real-time adaptability ensures that packets and delivery drivers alike are continuously redirected along the fastest available path.